KNECK OFFICIAL

Biology — Paper 3

231/3 • ORIGINAL REVISION SET 01 • 1 hour 45 minutes • 40 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Attempt all tasks. Use school laboratory supervision. Do not taste specimens or reagents. Supplied observations allow a clearly labelled simulated attempt when equipment is unavailable.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SUPPLIED MATERIAL / RESOURCE

Per student where available: iodine, Benedict reagent, dilute NaOH/CuSO4, heat-safe tubes, water bath, sample S prepared with glucose and egg albumen solution, soaked bean, hand lens, ruler and pencil. Teacher must manage allergies, hot water and chemical waste.

PRACTICAL / DATA TASKS

Three compulsory tasks.

1. Food tests on sample S: use separate portions. (a) Describe Benedict test with hot-water bath and result for reducing sugar. (4) (b) Describe iodine test and positive result. (3) (c) Describe Biuret test and positive result. (4) (d) Give one precaution. (1) Supplied observations for simulation: Benedict gives brick-red precipitate; iodine stays yellow-brown; Biuret turns purple. Infer composition. (2) [14 marks]

2. Investigate a bean seed. Teacher provides a soaked bean; alternatively use this supplied description: two large fleshy halves enclose a curved embryonic axis; small shoot lies between halves and a pointed root at lower end. (a) Draw a large labelled opened seed showing cotyledons, plumule, radicle and testa. (6) (b) State functions of cotyledon, testa and radicle. (3) (c) Plan an investigation showing water is needed for germination, including control and controlled variables. (5) [14 marks]

3. Leaf survey: quadrat counts for species X at 6 random sites are 4,7,5,8,6,0. Each quadrat is 0.5 m^2. (a) Calculate mean count, density and frequency percentage. (6) (b) Explain random placement and two limitations of using six quadrats. (3) (c) Plan one improvement and state two ethical precautions for fieldwork. (3) [12 marks]

END OF QUESTIONS

MARKING GUIDANCE

Biology • Paper 3 • Set 01

Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

PRACTICAL / DATA TASKS

1. (14 marks) Benedict plus sample, heat safely in hot-water bath, green/yellow/orange/brick-red precipitate indicates reducing sugar (4). Iodine blue-black if starch present; supplied negative (3). Add NaOH then a few drops dilute CuSO4; purple/lilac indicates protein (4). No direct flame on reagent tube/use holder/avoid corrosive contact (1). Supplied sample reducing sugar and protein present, starch not detected (2, require correct combined inference).

2. (14 marks) Clear proportional single-line drawing 2, four correct labels 4 (6); stored food, protection, embryonic root respectively (3). Same viable seeds, moist and dry groups, otherwise identical oxygen/temperature/light, replicates, record proportion germinated after fixed time (5). Description-based drawing labelled reconstruction, not direct observation.

3. (12 marks) Mean=30/6=5; density=10 plants/m^2; frequency=5/6×100=83.3% (6). Avoids selection bias; small sample/patchiness/seasonal variation (3). More spatially distributed random samples; do not damage organisms/habitats; obtain permission/avoid protected species (3).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf